Work
(6.4)

Suppose a force acts on an object which is moving in a line. The **work** is the energy transfered to the object by the force. If the force is **constant**, then the work equals

$$\text{force} \cdot \text{displacement}$$

The units of work are

SI: joules (J), i.e. $kg \cdot m^2/s^2$ or $N \cdot m$

US: foot-pound ($ft \cdot lb$)

An example of a constant force on an object is Earth's gravity. The weight of an object is the force of Earth's gravity on that object.

- In SI, usually the mass (kg) of an object is given, and we multiply by Earth's gravitational constant $g \approx 9.8\ m/s^2$ to get the weight in Newtons (N)
- In US, usually the weight (lb) of an object is given, so we don't multiply by g.

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward with a constant 12 lb of force, and it leaves my hand at 5 ft above the ground. The ball flies upward until it reaches 10 ft, then it falls to the ground.

How much work is needed to lift a 3 kg weight 2 m off the ground?

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull all the rope up the cliff?

A cylindrical tank with radius 2 m and height 8 m is filled with water, which has density 1000 kg/m^3. How much work is needed to pump all the water out of the top of the tank?

A conical tank with radius 2 m and height 8 m is filled with water, which has density 1000 kg/m^3. How much work is needed to pump all the water out of the top of the tank (the base of the cone)?

Suppose a force acts on an object which is moving in a line. The **work** is the energy transfered to the object by the force. If the force is **constant**, then the work equals

$$\text{force} \cdot \text{displacement}$$

Suppose the force is not constant, and is instead a function $F(x)$ of the position x. Then the work equals

$$\int_a^b F(x)\,dx$$

where a and b are start and end values of x.

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward with a constant 12 lb of force, and it leaves my hand at 5 ft above the ground.

Suppose instead the force is 0 at $y=3$, increases linearly until it is 12 lb at $y=4$, and then stays constant at 12 lb until $y=5$.

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull all the rope up the cliff?